PUSHFORWARDS AND HIGHER DIRECT IMAGES OF ADEQUATE MODULES ARE ADEQUATE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file adequate.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof's closing sentence states that the last case 'proceeds in exactly the same manner as in the paragraph above and is omitted'; this is the author's own named-omission of a repetition of the immediately preceding displayed induction step, and the sentence is delivered verbatim rather than summarised.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Computation of the cohomology of an object of a big site, and of sheaves on a site as cohomology of the small site. Topologies, Lemma topologies-lemma-morphism-big-fppf and Cohomology on Sites, Lemma sites-cohomology-lemma-cohomology-of-open, invoked at proof lines 2138-2146. Give the first and third equalities in the computation of R^i f_* F.
- Identification of the cohomology of an adequate module computed in the big and in the Zariski topology. Same chapter, Lemma lemma-same-cohomology-adequate, invoked at proof lines 2146 and 2156-2160. Gives the last equality and identifies F_B as an ordinary quasi-coherent sheaf.
- Adequacy of a functor is detected after sheafification. Same chapter, Lemma lemma-sheafification-adequate, invoked at proof lines 2147-2149. Reduces the claim to the displayed claim about the functor B -> H^q(T_B, F_B).
- Adequacy of the zeroth cohomology functor for affine T. Same chapter, Lemma lemma-pushforward-adequate, invoked at proof lines 2171-2172. Settles Case I.

Source reference. Original lemma 01EB: sources/stacks-chow/tags.txt, source lines 1828–1828. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06V2: sources/stacks-adequate/pinned-original-source.tex, source lines 684–684. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:adequate.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 06V8: sources/stacks-adequate/pinned-original-source.tex, source lines 910–910. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06V3: sources/stacks-adequate/pinned-original-source.tex, source lines 709–709. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06VA: sources/stacks-adequate/pinned-original-source.tex, source lines 968–968. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06VM: sources/stacks-adequate/pinned-original-source.tex, source lines 1896–1896. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06VP: sources/stacks-adequate/pinned-original-source.tex, source lines 1985–1985. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01KP: sources/stacks-chow/tags.txt, source lines 2058–2058. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03F3: sources/stacks-chow/tags.txt, source lines 4478–4478. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 021W: sources/stacks-chow/tags.txt, source lines 2685–2685. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $f : T \to S$ be a quasi-compact and quasi-separated morphism of schemes. For any adequate $\mathcal{O}_T$-module on $(Sch/T)_\tau$ the pushforward $f_*\mathcal{F}$ and the higher direct images $R^i f_*\mathcal{F}$ are adequate $\mathcal{O}_S$-modules on $(Sch/S)_\tau$.*

Proof. First we explain how to compute the higher direct images. Choose an injective resolution $\mathcal{F} \to \mathcal{I}^\bullet$. Then $R^i f_*\mathcal{F}$ is the ith cohomology sheaf of the complex $f_*\mathcal{I}^\bullet$. Hence $R^i f_*\mathcal{F}$ is the sheaf associated to the presheaf which associates to an

object U/S of $(Sch/S)_\tau$ the module

$$\begin{aligned}\frac{\mathrm{Ker}(f_*\mathcal{I}^i(U) \to f_*\mathcal{I}^{i+1}(U))}{\mathrm{Im}(f_*\mathcal{I}^{i-1}(U) \to f_*\mathcal{I}^i(U))} &= \frac{\mathrm{Ker}(\mathcal{I}^i(U \times_S T) \to \mathcal{I}^{i+1}(U \times_S T))}{\mathrm{Im}(\mathcal{I}^{i-1}(U \times_S T) \to \mathcal{I}^i(U \times_S T))} \\ &= H^i(U \times_S T, \mathcal{F}) \\ &= H^i((Sch/U \times_S T)_\tau, \mathcal{F}|_{(Sch/U\times_S T)_\tau}) \\ &= H^i(U \times_S T, \mathcal{F}|_{(U\times_S T)_{Zar}})\end{aligned}$$

The first equality by Topologies, Lemma 021W (and its analogues for other topologies), the second equality by definition of cohomology of $\mathcal{F}$ over an object of $(Sch/T)_\tau$, the third equality by Cohomology on Sites, Lemma 03F3, and the last equality by Lemma 06VM. Thus by Lemma 06VP it suffices to prove the claim stated in the following paragraph.

Let A be a ring. Let T be a scheme quasi-compact and quasi-separated over A. Let $\mathcal{F}$ be an adequate $\mathcal{O}_T$-module on $(Sch/T)_\tau$. For an A-algebra B set $T_B = T \times_{\mathrm{Spec}(A)} \mathrm{Spec}(B)$ and denote $\mathcal{F}_B = \mathcal{F}|_{(T_B)_{Zar}}$ the restriction of $\mathcal{F}$ to the small Zariski site of T_B. (Recall that this is a "usual" quasi-coherent sheaf on the scheme T_B, see Lemma 06VM.) Claim: The functor

$$B \longmapsto H^q(T_B, \mathcal{F}_B)$$

is adequate. We will prove the lemma by the usual procedure of cutting T into pieces.

Case I: T is affine. In this case the schemes T_B are all affine and $H^q(T_B, \mathcal{F}_B) = 0$ for all $q \geq 1$. The functor $B \mapsto H^0(T_B, \mathcal{F}_B)$ is adequate by Lemma 06VA.

Case II: T is separated. Let n be the minimal number of affines needed to cover T. We argue by induction on n. The base case is Case I. Choose an affine open covering $T = V_1 \cup \ldots \cup V_n$. Set $V = V_1 \cup \ldots \cup V_{n-1}$ and $U = V_n$. Observe that

$$U \cap V = (V_1 \cap V_n) \cup \ldots \cup (V_{n-1} \cap V_n)$$

is also a union of $n-1$ affine opens as T is separated, see Schemes, Lemma 01KP. Note that for each B the base changes U_B, V_B and $(U \cap V)_B = U_B \cap V_B$ behave in the same way. Hence we see that for each B we have a long exact sequence

$$0 \to H^0(T_B, \mathcal{F}_B) \to H^0(U_B, \mathcal{F}_B) \oplus H^0(V_B, \mathcal{F}_B) \to H^0((U\cap V)_B, \mathcal{F}_B) \to H^1(T_B, \mathcal{F}_B) \to \ldots$$

functorial in B, see Cohomology, Lemma 01EB. By induction hypothesis the functors $B \mapsto H^q(U_B, \mathcal{F}_B)$, $B \mapsto H^q(V_B, \mathcal{F}_B)$, and $B \mapsto H^q((U \cap V)_B, \mathcal{F}_B)$ are adequate. Using Lemmas 06V3 and 06V2 we see that our functor $B \mapsto H^q(T_B, \mathcal{F}_B)$ sits in the middle of a short exact sequence whose outer terms are adequate. Thus the claim follows from Lemma 06V8.

Case III: General quasi-compact and quasi-separated case. The proof is again by induction on the number n of affines needed to cover T. The base case $n = 1$ is Case I. Choose an affine open covering $T = V_1 \cup \ldots \cup V_n$. Set $V = V_1 \cup \ldots \cup V_{n-1}$ and $U = V_n$. Note that since T is quasi-separated $U \cap V$ is a quasi-compact open of an affine scheme, hence Case II applies to it. The rest of the argument proceeds in exactly the same manner as in the paragraph above and is omitted. □